
EE 502P: Analog System Design Laboratory

Course Code: EE 502P

Course Name: Analog System Design Laboratory

L-T-P-C: 0-0-3-2

Students intended for: EE 3rd year and CS 3rd Year

Elective or Core: Elective

Prerequisite: Knowledge of basic analog electronics, basic circuit analysis, Networks and Systems

Approval: 2nd Senate

The course is based on the Analog System Lab modules prepared by Texas Instruments. System Lab Kit ASLKv 2010 Starter Kit from Texas Instruments will be used for performing the experiments and also simulation tools will be used for analysis exhaustively.

Experiments:

- Negative Feedback Amplifiers and Instrumentation Amplifier
- Regenerative Feedback System, Astable and Monostable Multivibrator
- Integrators and Differentiators
- Analog Filters
- Self Tuned Filters
- Function Generator and Voltage Controlled Oscillator
- Phase Locked Loop
- Automatic Gain Control/Automatic Volume Control
- DC-DC Converter
- Low Dropout (LDO)/Linear Regulator

Text & Reference Books:

- Jerald G. Graeme, **Applications of Operational Amplifiers**, Third Generation Techniques
- James K. Roberge, **Operational Amplifiers: Theory and Practice**, Wiley.
- B Razavi, **Fundamentals of Microelectronics**
- A. Sedra and K. Smith, **Microelectronic Circuits**